



HIETANEN CONSULTANCY • CUSTOMER BRIEFING • SEPTEMBER 2026

FSharp.Azure.Quantum

Describe the problem. Skip the physics.

Quantum optimisation that ordinary developers and AI agents can ship. The problem goes in as everyday .NET code, not quantum circuits; the library builds and runs the circuits on a simulator or real quantum hardware and returns the answer. The algorithms and error correction are already done. First application: coordinating drone fleets.

hietanen.co.uk • github.com/Thorium/FSharp.Azure.Quantum

Coordinating a fleet is a combinatorial problem that grows faster than the fleet.



Routing and assignment are NP-hard

Which drone flies where, in what order, within range and airspace limits. Add one failed unit or a new task and the whole plan has to be recomputed, in seconds.



Quantum tooling is written for physicists

Qiskit, Cirq and Q# expect gates, Hamiltonians and hand-encoded QUBOs. A fleet-software team has none of that vocabulary and no way to hire it.



Hardware is arriving without a developer path

Azure Quantum, D-Wave and Braket sell machine time today. Nothing lets an ordinary .NET or C# application describe a business problem and run it there.

One NuGet package. Business problem in, quantum-computed answer out.

A .NET developer — or the agent working alongside them — describes routing, scheduling, assignment or portfolio problems in ordinary F# or C# and solves them on a simulator or on real hardware: the same code, every backend.

- On NuGet; F# computation expressions and a C# fluent API
- Seven optimisation builders, business builders for scheduling, fraud and churn, and quantum ML
- Backends: local simulator, Azure Quantum (IonQ, Pasqal, Quantinuum, Rigetti), D-Wave, AWS Braket, topological simulator
- OpenQASM 1.0–3.0 in and out, QIR emission, MAVLink missions; error mitigation and qubit routing

FleetPlanning.fs

```
let plan = tsp {  
    cities waypoints  
    distances haversine  
    minimize TotalFlightTime  
}  
let result =  
    HybridSolver.solve plan backend  
// splits the tour into range-safe  
// sorties across the whole fleet
```

Three layers. Developers touch only the top one.

Layer 1

Builders – the business vocabulary

TSP, scheduling, assignment, knapsack, portfolio, network flow; fraud, churn, workforce. Written by .NET developers or by their agents.



Layer 2

Solvers – the algorithms

QAOA for optimisation, VQE and ADAPT-VQE for chemistry, Grover, QFT-based arithmetic and phase estimation.



Layer 3

Backends – one interface, every machine

Local simulator, gate-based QPUs through Azure Quantum and Braket, D-Wave annealers, a topological simulator. Circuits arrive as OpenQASM from Qiskit and leave as OpenQASM, QIR or a MAVLink mission.

Quantum-first, honestly

A quantum solver never silently hands back a classical answer: if the hardware cannot solve it, the caller gets an error, not a disguise. HybridSolver routes to a classical baseline only when asked.

Four runnable fleet examples, all four flown in ArduPilot's own simulator.

FleetPathPlanning

Fleet path planning

Waypoints become a tour, split into range-safe sorties across the fleet, altitudes checked against the regulatory ceiling.

SwarmTaskAllocation

Swarm task allocation

Tasks with dependencies and priorities scheduled across drones with capacity limits, minimising mission time.

SwarmChoreography

Swarm choreography

Formation transitions solved with QAOA, exported to Crazyflie indoors and as ArduPilot missions outdoors.

FireAirBridge

Wildfire air bridge

Water as flow on corridors between lake and fire, capped by fill slots, spacing and drop slots. The planner opens corridors as the wind turns.



Where the hardware is. Every example exports ArduPilot missions and a launcher that checks, flies and draws them. All four have flown in ArduPilot's simulator (SITL): the real autopilot firmware on a simulated airframe. Next comes one real aircraft.

WHY NOW

The rules already allow swarms. The software is what is missing.

One pilot, many aircraft, already lawful

An EASA operational authorisation records a remote-pilot-to-aircraft ratio – 1:5, or a pool of three pilots to ten aircraft. The cap comes from the automation the aircraft's flight manual documents, not from a rule still to be written.

Deployment, not legislation, is the bottleneck

UK BVLOS flies today in atypical air environments – rail, power-line and wind-farm corridors – under a handful of authorisations. Since the EU's U-space rules applied in 2023, one U-space airspace is live.

Quantum hardware is on the cloud

Azure Quantum sells IonQ, Pasqal, Quantinuum and Rigetti time; D-Wave annealers handle thousands of variables; Microsoft is building the Majorana machine we already simulate.

AI agents need a typed API

Coding agents can write a builder expression; they cannot invent a QUBO encoding. A high-level, typed library is how quantum reaches the code that agents will write.

So the developer layer arrives **before** the fleets – which is the only time it can be built, and the reason the buyer today is the vendor shipping the fleet software rather than the operator flying it. The revenue does not wait for quantum: what a customer buys is fleet planning that works now on classical hardware, with the quantum path wired behind the same API.

COMPETITION

Physicist SDKs on one side, classical optimisers on the other. We sit between.

Capability	FSharp.Azure.Quantum	Quantum SDKs	Classical optimisers	Quantum startups
Business-level problem builders	Yes	No, gates and circuits	Yes, modelling languages	Per engagement
Usable from .NET and C#	Yes	Python, Q# only	Yes	Rarely
Runs on every quantum backend	Yes, one interface	Own vendor mostly	No	Own stack
Solves large fleets today	Hybrid path	No	Yes, the benchmark	Claimed
Evidence for a multi-aircraft authorisation	Constraint audit trail	No	No	Per engagement
Drone flight-controller export	Built in	No	No	No
Classical fallback disclosed	Always	Not applicable	Not applicable	Often not
Circuits move in and out	OpenQASM 1.0–3.0, QIR	Yes, OpenQASM	Not applicable	Own format

Quantum SDKs: Qiskit, Cirq, Q#, D-Wave Ocean. Classical optimisers: Gurobi, OR-Tools, CPLEX. Quantum startups: application-specific consultancies built on one vendor.

One engineer, AI-assisted, shipped the library. The company grows as vendors sign.



Tuomas Hietanen – Founder and CEO

CTO and co-founder of Finular Group, nine years building regulated lending platforms in the UK and US. Lead engineer at Basware on purchase-to-pay at global scale. Maintainer of SQLProvider and LINQKit, published .NET author, and the author of FSharp.Azure.Quantum.

How the work gets done

One engineer directing AI coding agents, a test suite in the thousands, the source public – and a network of engineers to call on, .NET engineers from Ukraine among them.

Next hires

An engineer when two vendors ship the planner; a quantum-algorithms researcher and a head of sales as projects repeat.

Key-person risk, said out loud. One person holds the code; delivery already draws on the network. The code is public domain and under test, and every algorithm is documented against its source paper.

WORKING WITH US

Bring us a fleet-planning problem.

01 • Project

Twelve weeks, fixed scope

Your fleet or your product, your constraints, a planner in production.

02 • Supported builds

For vendors who ship it

Signed builds, long-term support, an SLA and private builders.

03 • What stays yours

Aircraft and authority

Every mission is a file your pilot-in-command inspects before anything flies.

Ask for a live demo: four drones, one laptop, a quantum solver, twenty minutes.

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